Siddharth Karanam

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Education

Vellore Institute of Technology Aug 2023 – May 2027
B.Tech in Computer Science and Engineering *CGPA: 8.46*

Achievements

Pentathon 2024 Finalist | *NCIIPC–AICTE National Cybersecurity Competition* 2024

- Selected as a finalist in a **national-level cybersecurity** competition focused on secure systems, threat analysis, and **VAPT.**

Open Source Contributions

Linux Kernel | *C, Device Tree, TI K3 remoteproc, TI-SCI* | Patch 1, 2, 3 2026

- Authored a patch series adding a **ti_sci_dev_ops abstraction** (`ti_sci_dev.h`) to maintain a **single source of origin** for all `ti_sci` device ops call sites (`is_on`, `get_device`, `put_device`, and the `ti,sci-dev-id` property read) across the **TI K3 R5, M4, and DSP remoteproc drivers**.
- Replaced `devm_kcalloc` with `devm_kzalloc` for the **buffer pointer allocation** in the **TI-SCI firmware driver**, since a single-element pointer does not need array multiplication overflow checks.

jagua-rs | *Rust, WebAssembly* | PR #47 2025

- Added **WebAssembly** build support for a Rust-based **computational geometry** library.
- Refactored core logic to be **WASM-compatible** by removing **platform-specific assumptions**.
- Implemented a **wasm-bindgen interface** to expose library functionality to **JavaScript runtimes**.
- Ensured **deterministic behavior** across **native** and **WebAssembly** targets.

SWHKD | *Rust, Wayland, libudev* | PR #306, #304, #302 2025

- Optimized internal data handling by reducing **allocations** and improving **ownership semantics**.
- Refactored **environment** and **configuration logic** using allocation-aware Rust patterns.
- Improved **error handling**, **logging clarity**, and **configuration performance**.

Hyprland | *C++, Wayland Compositor* | PR #10289, #10246 2025

- Refactored compositor utilities using **std::string_view**, **std::optional**, and **std::filesystem**.
- Modernized desktop-related code using **std::ranges** and cleaner **class-scoped APIs**.
- Improved **code safety**, **readability**, and consistency across **system utilities**.

Experience

Research Commons | *C++ Developer Intern* May 2025 – Nov 2025

- Worked with **concurrency primitives** and **memory ordering** considerations in multi-threaded systems.
- Contributed to a custom **tensor** and **numerical computation** library implemented in modern **C++**.
- Worked on low-level components involving **memory ownership**, **cache-aware data layouts**, and **threading**.
- Performed **heap and CPU profiling**, **micro-benchmarking**, and performance analysis on **Linux**.
- Assisted in improving **execution pipelines** and debugging **system-level performance regressions**.

PostgreSQL Proxy Server | *Rust Developer (Freelance)* Oct 2024 – Dec 2024

- Designed **SQL query transformation** logic using **AST manipulation** for a PostgreSQL proxy layer.
- Improved **SQL parsing and validation** using **sqlparser-rs** for stricter schema enforcement.
- Worked on **metadata** and **information schema** handling for proxy-side query inspection.
- Improved **logging**, **error handling**, and **test coverage** for production debugging.

Projects

inLimbo | *C++, ALSA, FFmpeg (libav), DBus/MPRIS* 2025

- Implemented a standalone **inLimbo-core** handling **song indexing**, **serialization**, **caching**, and the **audio engine**.
- Designed an **intuitive, stable API** enabling various frontends without leaking backend internals.
- Uses Linux native **ALSA + FFmpeg (libav)** for precise, low-level Linux audio sampling and control.
- Added **gapless playback**, **instant seeks**, and **low-latency metadata access** without sacrificing performance.
- Integrated **MPRIS over DBus** with extensible backends for seamless system-wide media control.
- Built a **plugin-based frontend system** with a stable ABI to allow independent frontend development.

Wavy | *C++, FFmpeg, OpenSSL, oneTBB* 2025

- Designed and implemented a **modular LAN audio streaming system** with **runtime-loaded plugin architecture**.
- Integrated **FFmpeg (libav)** for audio decoding, normalization, and **real-time streaming**.
- Implemented **encrypted**, **compressed**, and **archivable** audio transport using **OpenSSL**.
- Designed **plugin-based frontends and audio backends** around a shared core.
- Built a **resilient, cloud-deployable server** with low-latency metadata queries.

obZcene | *C, SDL2, SIMD* 2025

- Developed a **software rasterizer** from scratch in **pure C** using **SDL2**
- Implemented a **math library functions** optimized with **architecture-specific SIMD intrinsics** (x86)
- Features a **Blender-exported OBJ loader** with **texture and material rendering**
- Designed an **intuitive and modular rendering API** to simplify **game and engine development**

Anvilock | *C, Wayland, PAM, EGL* 2024

- Developed a secure **Wayland session locker** with **PAM-based authentication**.
- Implemented **EGL-based rendering** compliant with **ext-session-lock-v1** for compositors.
- Designed an **asynchronous resource pipeline** to enable extensible script friendly widgets.
- Built efficient **shader and rendering state management** using strict **RAII-style lifetime control**.

tibs | *C++, CLI11, nlohmann-json* 2026

- Built **tibs**, a **C++23** project & package manager that brings **Cargo-like ergonomics** to native C/C++ builds: projects are written in a declarative DSL and transpiled to **CMake/Ninja**, so the tool *drives* existing build systems instead of replacing them.
- Engineered the underlying **melon** transpiler as a full **compiler frontend** (lex → parse → semantic analysis → build-graph IR → CMake/Ninja backends) with rich diagnostics, plus a **dependency resolver** with a content-addressed cache, lockfile, transitive/cycle handling, and offline vendoring — eliminating CMake `FetchContent`.
- Shipped a cohesive **CLI** spanning project scaffolding, build/run/test lifecycle, dependency & cache management, and toolchain health checks.

Technical Skills

Languages: C, C++, ASM (x86), Rust, Zig, Bash, Java, Python, JavaScript
Systems & OS: Linux, POSIX, X11, Wayland, DBus, PAM, PipeWire, ALSA, WebAssembly
Profiling & Debugging: perf, gdb, Valgrind, heaptrack, strace, ltrace
Libraries & Tooling: FFmpeg (libav), LLVM, ncurses, OpenSSL, SDL, OpenGL/EGL, Boost C++, GLib
Build & Dev Tools: Git, Make, CMake, Meson, Clang, GCC, Docker
Databases: PostgreSQL, MySQL